signal and image proccecing lab

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Lab 6

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Q1

Analyze the given image (t1.jpg) by reading it using the `imread` command. Display the image using `imshow`, select the first slice of the image for processing, and compute the 1D Discrete Fourier Transform (DFT) of the 128th row. Plot the magnitude and phase of the DFT. Then, convert the image to a double format and compute the 2D FFT. Visualize the results with and without the `fftshift` for better interpretation.

solution

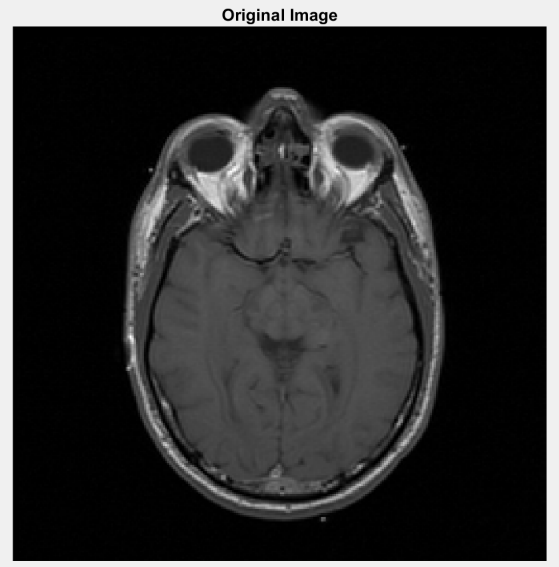

Figure 1. Original image

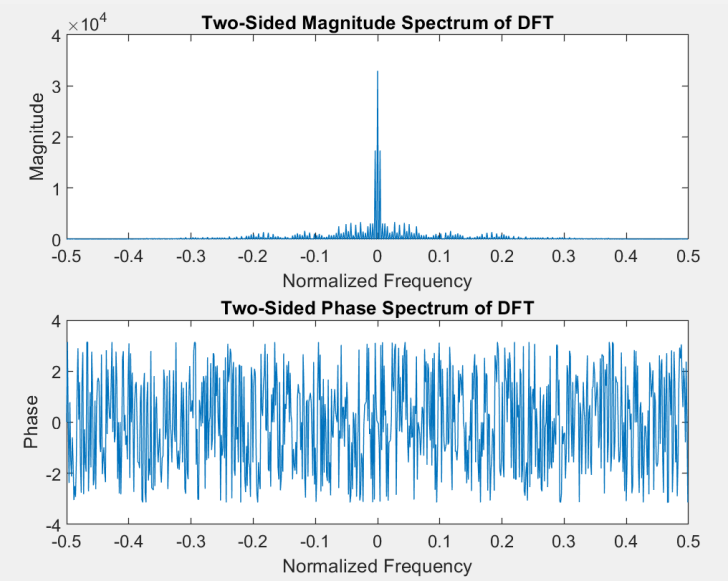

Figure 2. Magnitude and Phase

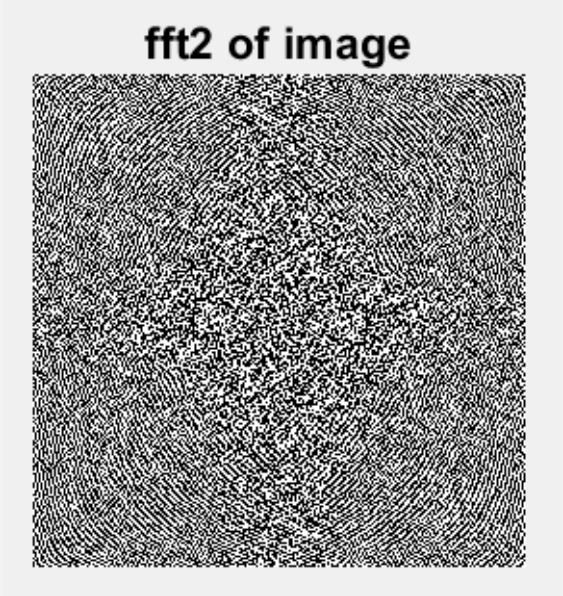

Figure 3. fft2

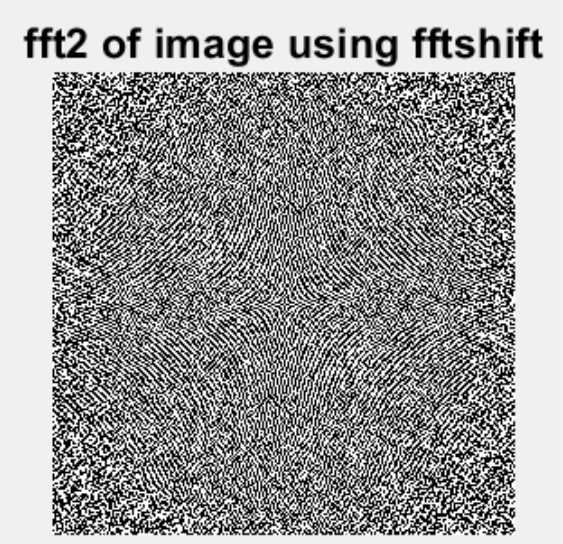

Figure 4. using fftshift

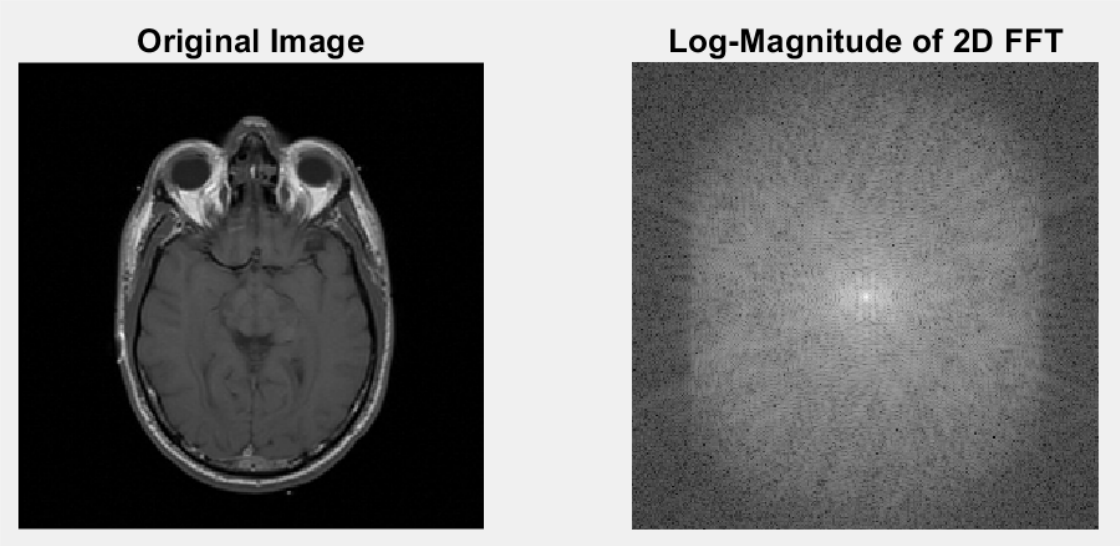

Figure 5. log of fft2 beside the original image

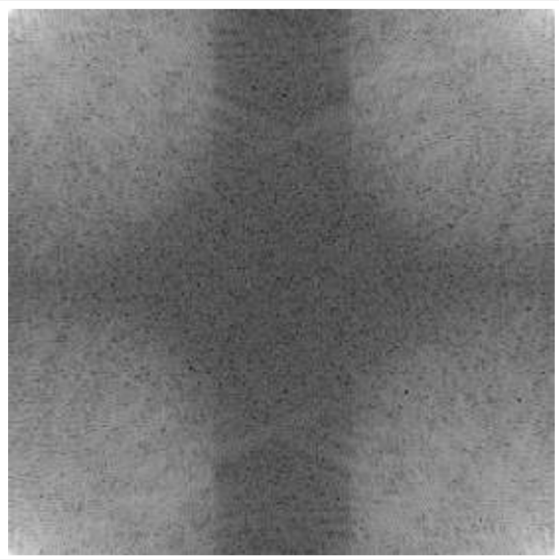

Figure 6. without fftshift

As we can see in the results above, the Magnitude of the spectrum is symmetric at zero point. It has the most amplitude at zero point. After some frequencies, we can see that we have strong attenuation. We can say that our changes are low in frequency. The 128th row has information about frequency components. The fftshift function repositions the zero-frequency component to the center, enhancing the clarity of the visualization. Without applying fftshift, the zero-frequency component remains in the corners, making the frequency spectrum harder to interpret intuitively.

Q2

Perform convolution of two matrices, G and L, using Fourier Transform and visualize the results. Then, apply convolution between a given image and G, and display the results for analysis.

solution

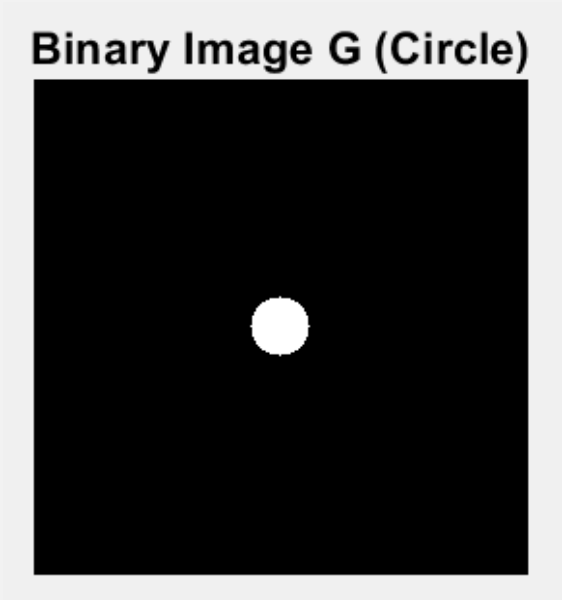

Figure 7. Circle G

Figure 8. F

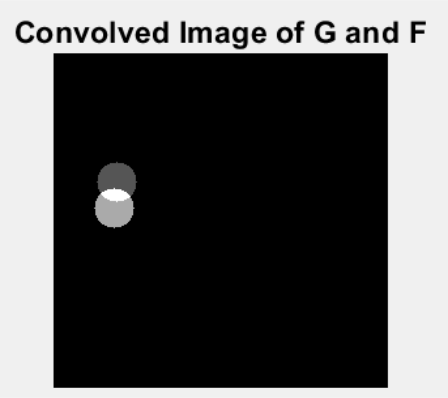

Figure 9. Convolution G and F

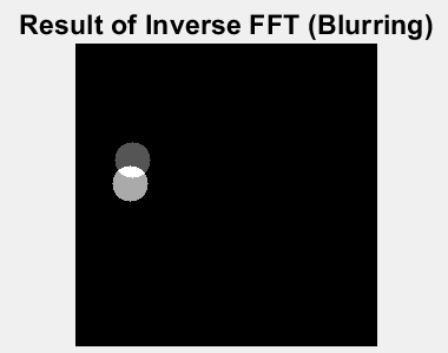

Figure 10. Result of inverse fft (blurring)

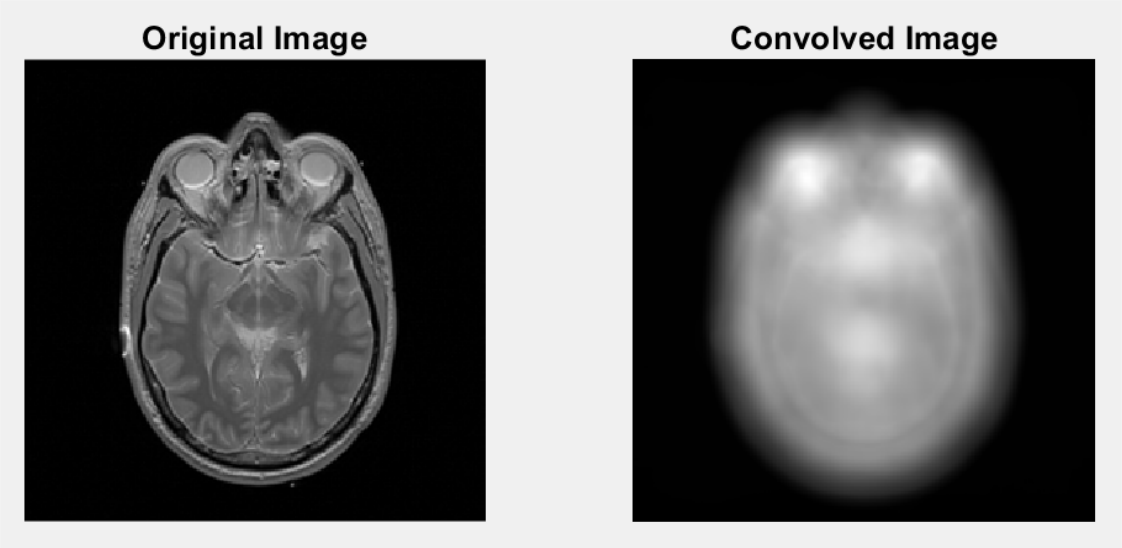

Figure 11. Original image and convolved image

In this part of the experiment, the convolution of two matrices, G and L, was performed by utilizing the Fourier Transform. Convolution in the frequency domain is equivalent to the multiplication of their respective Fourier transforms, which allows for an efficient computation. The result of this convolution exhibits a circular pattern centered at the positions defined by matrix L, with the intensity of the result depending on the values at those central positions. This outcome is in line with the characteristics of G, a low-pass filter that allows low-frequency components of L to pass through while suppressing higher frequencies.

When this convolution operation was applied to the given input image (pd.jpg), the filter G acted as a smoothing operator, emphasizing low-frequency features and effectively blurring the image. As a low-pass filter, G eliminates high-frequency components, such as fine edges and noise, while preserving the broader structures and patterns of the image. This filtering behavior is a direct consequence of the frequency-domain characteristics of G, which limits the spatial result to the circular region defined by the filter's radius.

The output image after applying the convolution clearly shows a reduction in sharp edges and fine details, resulting in a blurred effect. This is reflected in the second figure, where the image boundaries appear significantly softened. Such results are typical for low-pass filtering, which smooths the image both in the spatial domain and the frequency domain by restricting high-frequency information.

Overall, this experiment demonstrates the effectiveness of Fourier Transform-based convolution in performing low-pass filtering, highlighting how frequency-domain operations can be utilized for image smoothing and noise reduction.

Q3

Use the concept of zero-padding in the frequency domain to magnify the spatial domain representation of an image. The process involves padding the frequency representation of the image with zeros to increase its resolution and then performing an inverse Fourier transform to magnify the image.

solution

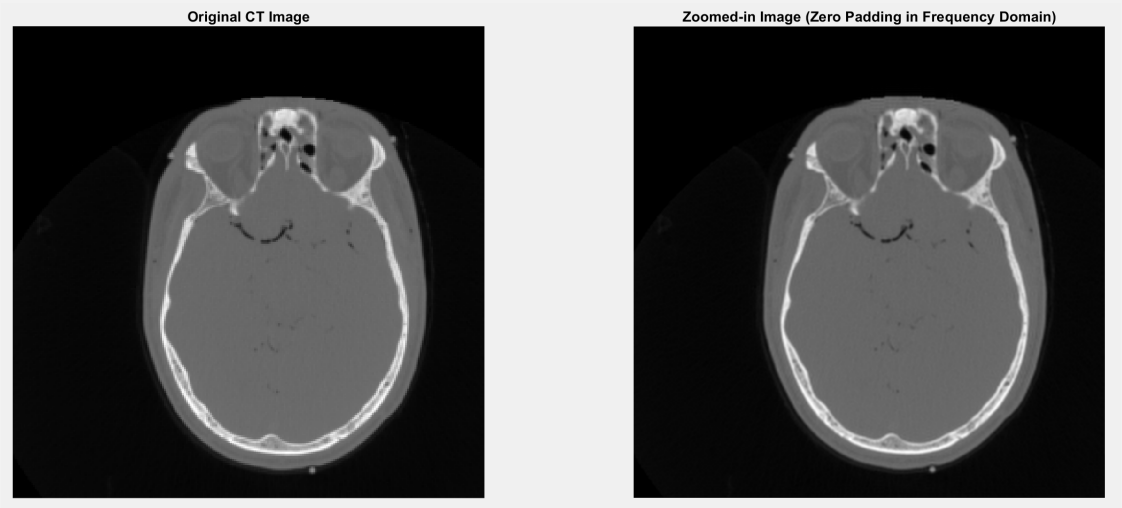

Figure 12. Original image and zoomed image

try to make the output better :

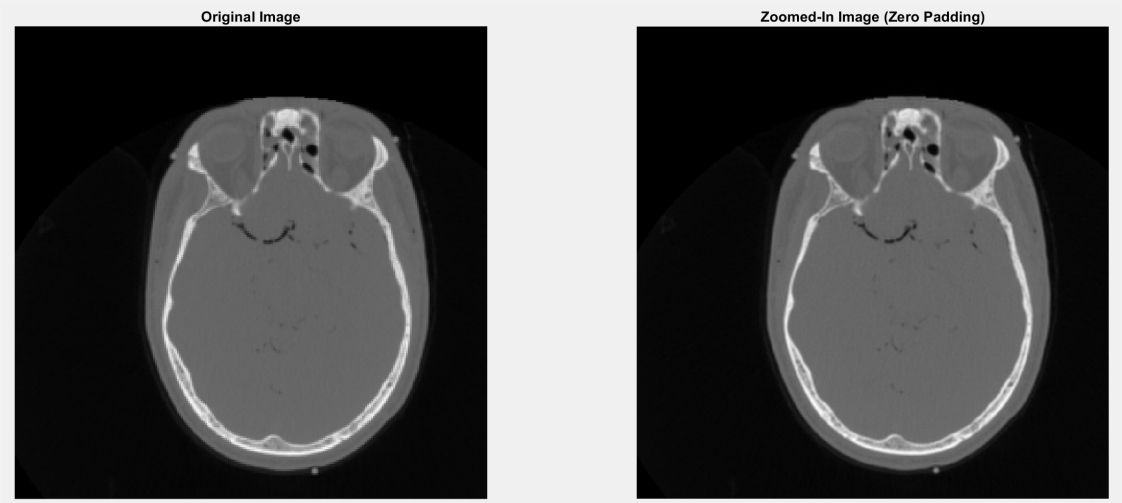

Figure 13. Original image and zoomed image

The technique of zero-padding in the frequency domain is employed to enhance the resolution of the spatial domain image. By adding zeros to the frequency representation of the image, we effectively increase the sampling density in the spatial domain after performing an inverse Fourier transform. This process allows us to magnify the image without adding any new information or fine details; it only interpolates existing data to produce a smoother, higher-resolution result.

Zero-padding introduces additional frequency components that correspond to the added zero values in the frequency domain. When the inverse Fourier transform is applied, these extra components cause the spatial image to appear enlarged. However, it's important to note that this enlargement doesn't introduce new features or sharpness into the image; it simply "zooms in" on the original image, providing a higher resolution version of it.

Zero-padding is an effective method for magnifying the image by manipulating its frequency representation. While the process increases the resolution, it does not add new content to the image, making it a useful technique for image zooming and resolution enhancement without altering the underlying structure.

Q4.1

The goal of this question is to create a spatial shift for the original image using processing in the Fourier domain and then return to the spatial domain.

Read the image `ct.jpg` from the folder `S1_Q4_utils`, and consider the first slice of the original image. Then, perform a spatial shift by 20 units to the right and 40 units downward.

Carefully perform this operation in the Fourier domain and then return to the spatial domain. Finally, plot the shifted Fourier kernel and analyze the result.

solution

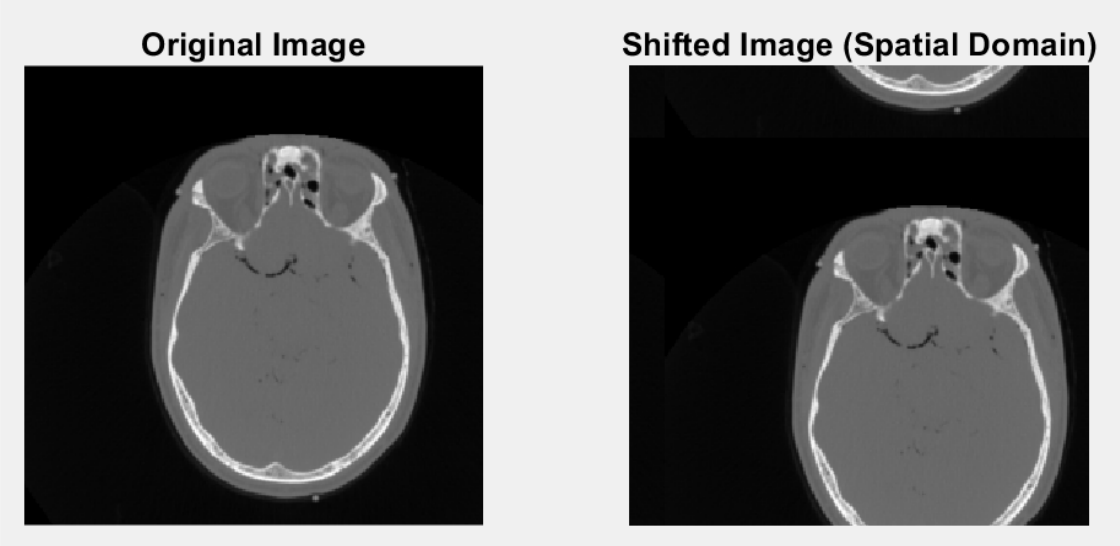

Figure 14. Original image and shifted image

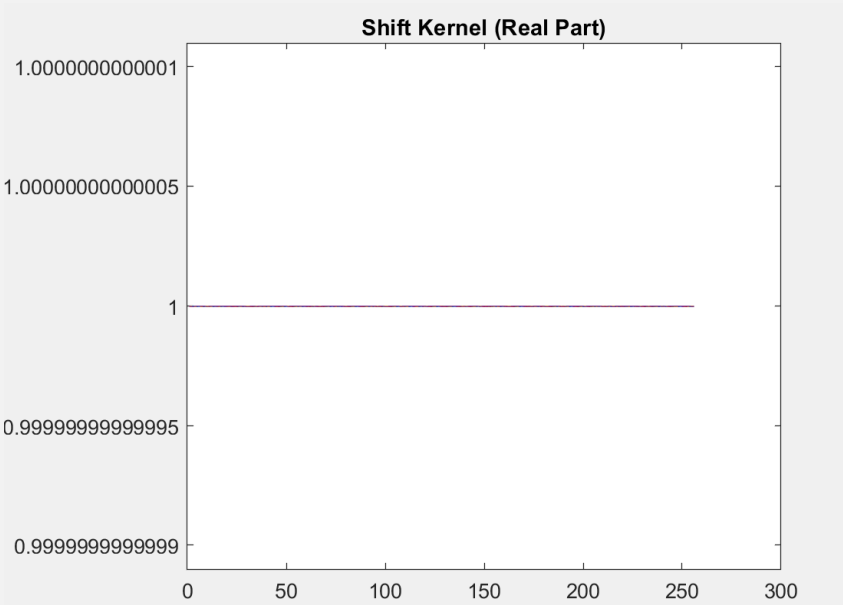

Figure 15. Shift kernel

The 2D Fourier Transform offers an insightful approach to understanding spatial transformations, allowing for efficient manipulation of an image in the frequency domain before returning it to the spatial domain.
In the Fourier domain, the image is represented as a collection of coefficients, a_k, corresponding to various spatial frequencies. These coefficients are organized in the 2D frequency space, with the horizontal and vertical axes, u_k and v_k, representing the respective spatial frequency components along the image's x and y axes.
These frequency-domain coefficients are crucial because they form the sinusoidal basis functions that reconstruct the image back into the spatial domain. The magnitudes and phases of these coefficients define the structure and features of the original image. By manipulating these coefficients, we can achieve various spatial transformations, such as shifting the image in both the horizontal and vertical directions.
In this experiment, the spatial shift was applied by modifying the frequency-domain representation. A shift of 20 units to the right and 40 units downward was introduced, and the inverse Fourier transform was then used to return to the spatial domain. The frequency-domain shift corresponds to a phase shift in the Fourier coefficients, which results in the translation of the image in the spatial domain.
The resulting shifted Fourier kernel, visualized in the figure, shows distinctive diagonal lines. These lines are indicative of the two-directional spatial shift. The orientation of the lines relative to the frequency axes reveals the direction and magnitude of the shift applied to the image. Specifically, the diagonal nature of the lines reflects the simultaneous shift in both the horizontal and vertical directions.
In conclusion, this experiment demonstrates how spatial shifts in an image can be achieved through careful manipulation of its Fourier transform. The frequency-domain approach provides an elegant way to perform such transformations, offering a clear view of the shift's impact on the frequency components.

Q4.2

Read the image `ct.jpg` from the folder `S1_Q4_utils` and consider the first slice of it.

Rotate the image by 30 degrees using the `imrotate` function. Plot the original image and the rotated image side by side in the spatial domain.

Compute the Fourier Transform of both the original image and the rotated image. Plot their Fourier Transforms side by side in the frequency domain.

Finally, display the rotated image in the frequency domain (its Fourier Transform) and analyze the results.

solution

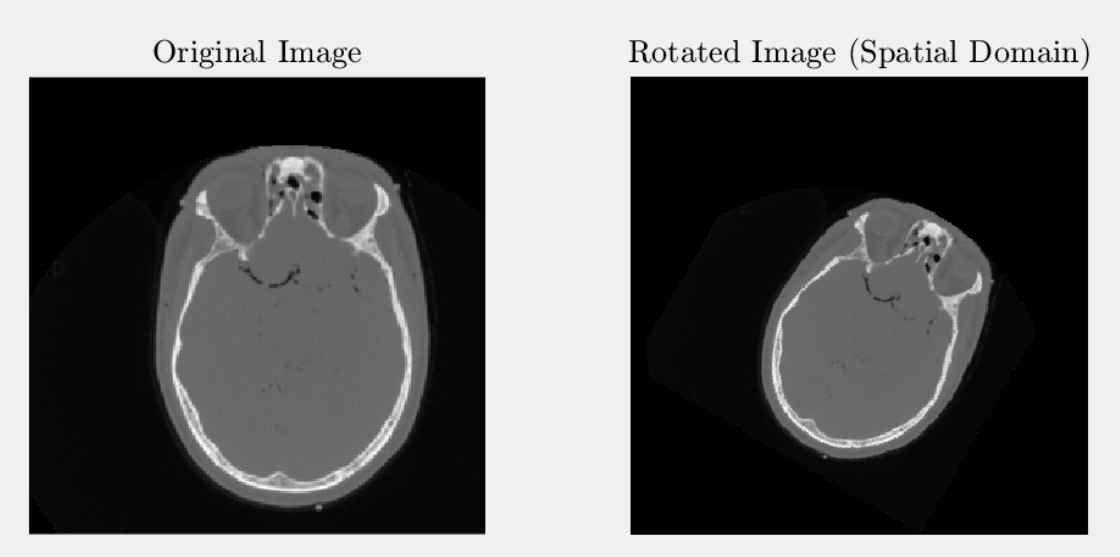

Figure 16. Original image and Rotated image

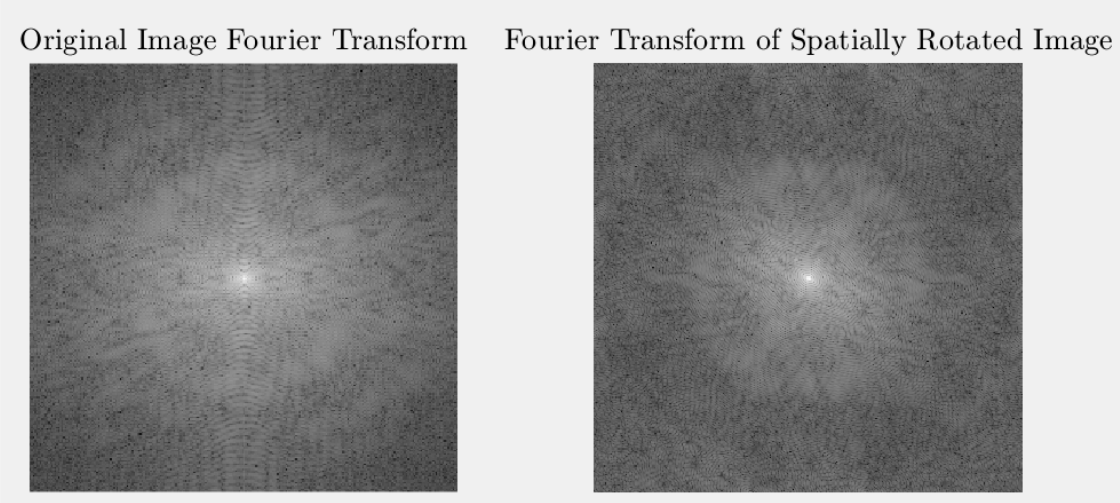

Figure 17. Fourier transforms of original and rotated images

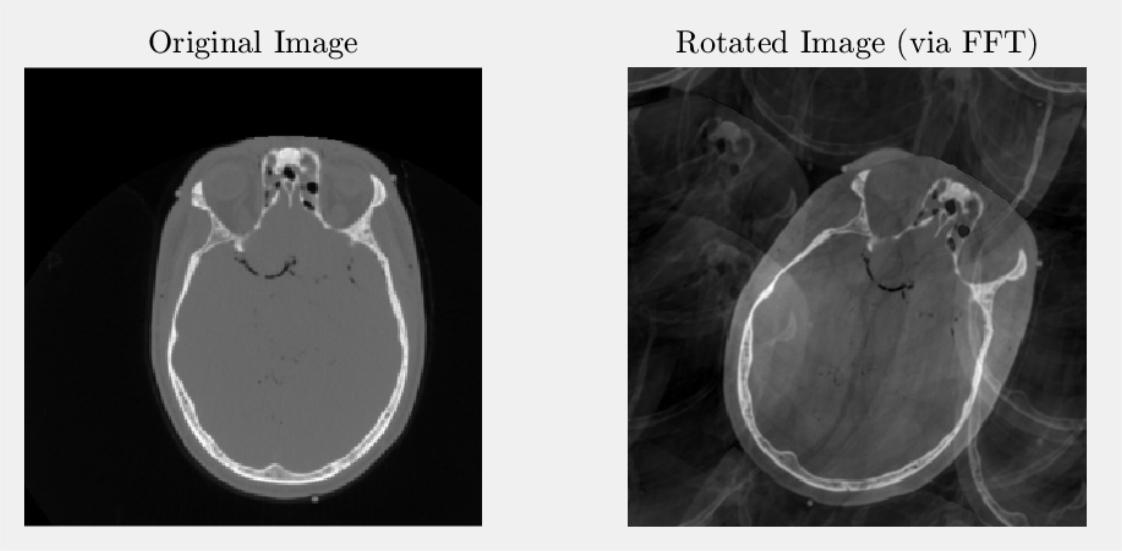

Figure 18. Original image and rotated image(via fft)

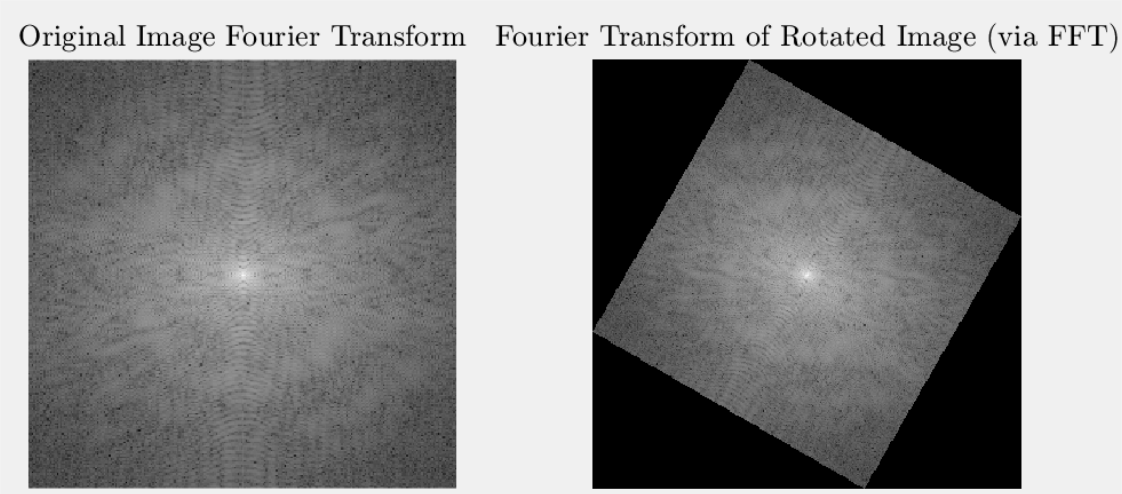

Figure 19. Original image and Fourier transform of rotated image

- The first image shows the original CT scan of the head in the spatial domain, before any transformations are applied.
- The second image displays the same CT scan, but it has been rotated by 30 degrees using spatial transformation techniques. The rotation preserves the intensity values of the pixels but alters their orientation.

Analysis of the Spatial Domain Images:

- Rotating the image introduces a geometrical change in its spatial layout. While the intensity values of the pixels remain unchanged, their relative positions are modified, resulting in a new orientation of the image.
- This transformation demonstrates that spatial operations like rotation affect the positioning of the image content but do not alter the inherent information contained within the image.

Description of the Frequency Domain Representations:

- The left image presents the Fourier Transform of the original CT scan, showing its frequency domain representation.
- The right image shows the Fourier Transform of the rotated CT scan, revealing the frequency components after the image rotation.

Analysis of the Frequency Domain Images:

- The Fourier Transform of the original image exhibits a symmetrical pattern, with low-frequency components concentrated at the center and high-frequency components spread towards the edges. This is typical of natural images, where most of the information lies in the low-frequency components.
- After the image is rotated, the Fourier Transform shows the same frequency components but with a corresponding rotation in the frequency domain. This rotation mirrors the spatial transformation, demonstrating that a rotation in the spatial domain is represented by an equivalent rotation in the frequency domain.
- This property aligns with the theoretical principles of the Fourier Transform: a spatial rotation leads to an identical rotation in the frequency domain, preserving the structure of the frequency components but altering their orientation.

The experiment highlights the relationship between the spatial and frequency domains. Rotating an image in the spatial domain results in a matching rotation of its frequency domain representation. This showcases how geometric transformations in the spatial domain have predictable and consistent effects on the Fourier Transformed data, which is an essential concept in signal processing and image analysis.

Q5

The goal of this task is to produce results similar to Figure 4-1. To this end, read the image `t1.jpg` from the folder `S1_Q5_utils`. Then, using the `circshift` function, compute the central difference derivatives in both vertical and horizontal directions.

Next, calculate the gradient magnitude for the image and display all the results, including:

- Original image,
- Vertical derivative,
- Horizontal derivative,
- Gradient magnitude.

Finally, analyze the results in terms of detecting horizontal, vertical, and diagonal edges alongside the original image.

solution

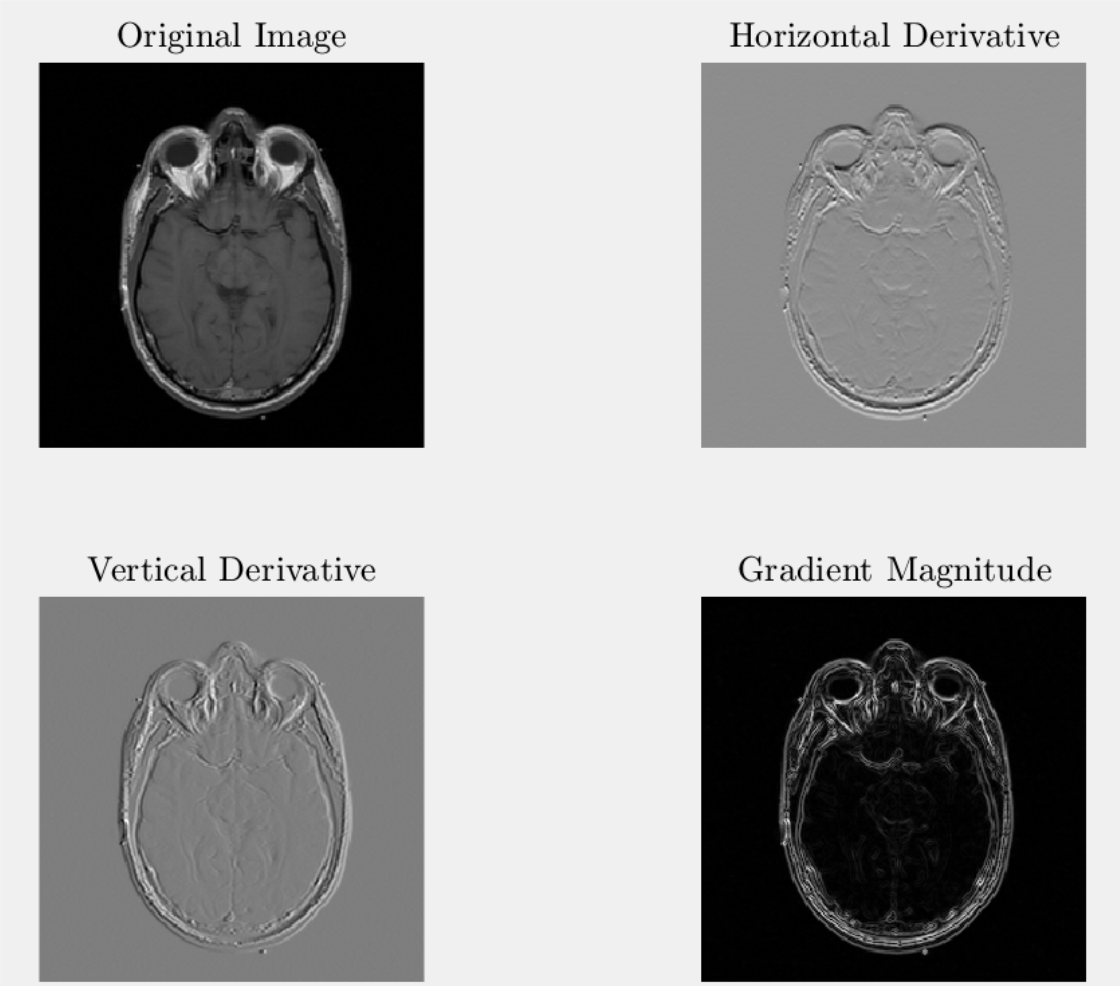

Figure 20. Original, Horizontal, and vertical derivatives and gradient magnitude

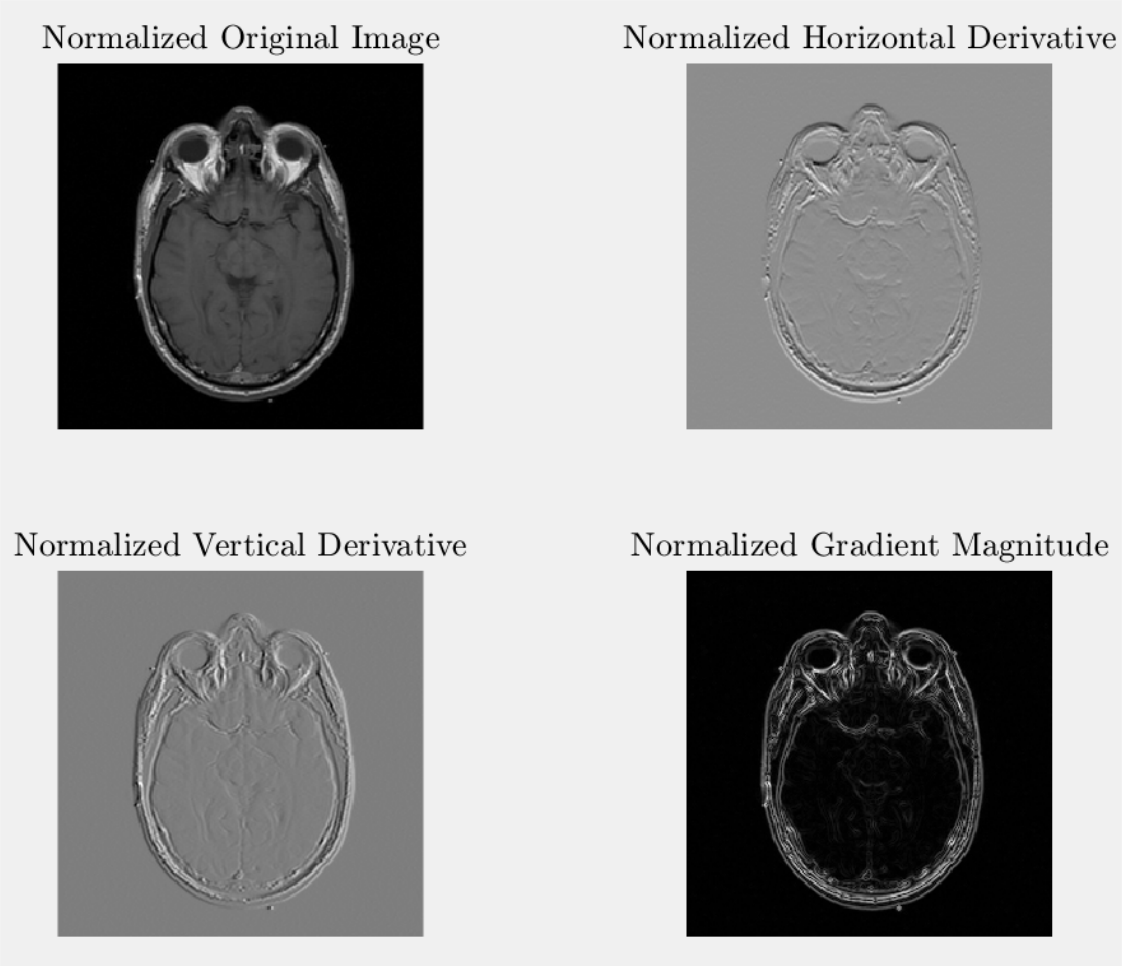

Figure 21. Normalized

- The **Original Image** represents the grayscale CT scan of the brain, capturing important structural details and anatomical features. This image serves as the foundation for subsequent edge detection.
- The **Vertical Derivative (dy)** focuses on identifying edges in the vertical direction. It highlights the presence of horizontal features, such as the outlines of the brain's structures and boundaries, by computing the change in intensity along the vertical axis.
- The **Horizontal Derivative (dx)** emphasizes the horizontal edges within the image. It reveals vertical structures, emphasizing features that change intensity across the horizontal direction, such as the separation between adjacent regions of the brain.
- The **Gradient Magnitude** is derived by combining both the vertical and horizontal derivatives. This result provides a complete representation of edges in the image, capturing structural transitions in all directions, including diagonal edges. The gradient magnitude visualizes the overall edge strength across the image.
- Together, these components demonstrate how derivative operations, along with gradient magnitude computation, can effectively highlight edges and boundaries, providing a clearer representation of structural features in medical images.

The process of calculating the derivatives and gradient magnitude provides a detailed analysis of the image's edges in various orientations. The vertical and horizontal derivatives reveal edges in their respective directions, highlighting significant boundaries in the image. The gradient magnitude serves as a comprehensive edge map, capturing both horizontal, vertical, and diagonal edges, which are crucial for analyzing the fine details of the brain's structure.
In medical image processing, such edge detection techniques are particularly useful for delineating anatomical structures and identifying features that may not be immediately apparent in the raw image. By combining vertical and horizontal derivatives, the gradient magnitude enables the detection of edges from all directions, thus offering a holistic view of the image's structural boundaries.
This approach underscores the effectiveness of derivative-based edge detection in emphasizing key structural elements within medical images, aiding in tasks such as segmentation, feature extraction, and image analysis.

Q6

Investigate advanced image edge detection methods, such as the **Canny** and **Sobel** methods. Apply these methods to the image from the previous question and analyze the results.

Finally, compare the results of these methods with the central difference method used earlier and discuss the differences.

solution

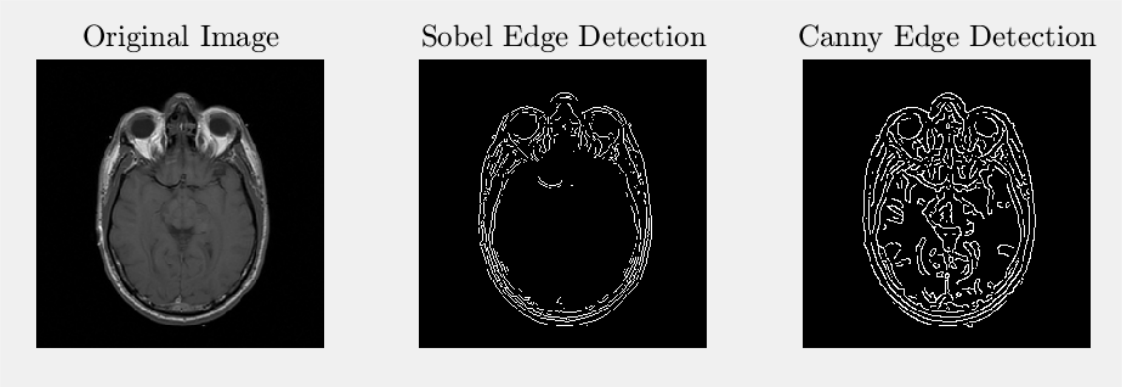

Figure 22. Sobel and Canny edge detection

Sobel Edge Detection:
The Sobel edge detection technique operates by calculating the gradients of an image in both the horizontal and vertical directions. This approach emphasizes areas where there is a rapid change in pixel intensity, which typically corresponds to edges. The Sobel method uses two convolution kernels for gradient computation:

$$G_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}, \quad G_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

These kernels highlight edges by computing a weighted gradient, capturing regions of high spatial frequency. The Sobel operator is computationally efficient, making it suitable for detecting prominent edges in an image. However, it may be less sensitive to fine details and can be affected by noise, as it does not inherently include noise reduction steps.

Canny Edge Detection:
The Canny edge detection method is a more advanced and robust approach, comprising several stages to ensure more precise edge detection. The process includes:

1. **Smoothing:** The image is initially smoothed using a Gaussian filter to reduce noise, making the subsequent edge detection steps more reliable.
2. **Non-Maximum Suppression:** This step refines the edges by retaining only the local maxima of the gradient, removing spurious or weak edges that do not correspond to true boundaries.
3. **Hysteresis Thresholding:** Two thresholds are used to distinguish strong and weak edges. Weak edges that are connected to strong edges are preserved, while isolated weak edges are discarded.
4. **Gradient Calculation:** The gradient magnitude and direction are computed using finite differences, which helps identify the most significant transitions in pixel intensity.

The Canny method is generally more reliable than Sobel, as it combines noise suppression with a multi-stage approach for edge detection. This makes Canny particularly effective in identifying well-defined edges while minimizing the impact of noise.

Comparison of Edge Detection Methods:

- The **Sobel** method is quick and efficient for detecting strong edges. However, it tends to be less effective in the presence of noise and may not capture fine details well. It is well-suited for detecting larger, more prominent edges.
- The **Canny** method, on the other hand, produces cleaner and more precise edges by incorporating steps such as noise reduction, non-maximum suppression, and hysteresis thresholding. This makes Canny highly robust, especially when dealing with noisy images or requiring finer edge details.
- When compared to the **Central Difference** method, both Sobel and Canny offer more structured and reliable results for edge detection. The Central Difference method captures basic edge information, but Sobel and Canny methods are more refined, with Canny being the most advanced for precise edge detection.

In this task, the advanced edge detection methods, Sobel and Canny, were applied to the image. While the Sobel operator is simple and efficient, it is outperformed by the Canny method, which offers superior edge detection accuracy through noise reduction and sophisticated edge-preserving techniques. Both Sobel and Canny provide more reliable edge detection compared to the central difference method, especially for capturing fine details and reducing the influence of noise in the image.